## Intersection of 3 Planes

**8.** Using Gaussian elimination, state whether the system has no solutions, a unique solution, or an infinite number of solutions.

a. $\begin{aligned} x + 2y + z &= 12 \\ 2x - y + z &= 5 \\ 3x + y - 2z &= 1 \end{aligned}$

b. $\begin{aligned} x - y + 2z &= 4 \\ 2x - 2y + 4z &= 7 \\ 3x - 3y + 6z &= 11 \end{aligned}$

c. $\begin{aligned} x + y - z &= 5 \\ 2x + 2y - 4z &= 6 \\ x + y - 2z &= 3 \end{aligned}$

d. $\begin{aligned} -2x + 4y + 6z &= -2 \\ 4x - 8y - 12z &= 4 \\ x - 2y - 3z &= 1 \end{aligned}$

e. $\begin{aligned} x + y + 2z &= 2 \\ x - y - 2z &= 5 \\ 3x + 3y + 6z &= 5 \end{aligned}$

f. $\begin{aligned} x + 3y + 5z &= 10 \\ 2x + 6y + 10z &= 18 \\ x + 3y + 5z &= 9 \end{aligned}$

g. $\begin{aligned} x - 3y - 2z &= 9 \\ x + 11y + 5z &= -5 \\ 2x + 8y + 3z &= 4 \end{aligned}$

h. $\begin{aligned} x + y + 2z &= 6 \\ x - y - 4z &= -2 \\ 3x + 5y + 12z &= 27 \end{aligned}$

i. $\begin{aligned} 2x + y + z &= 0 \\ x - 2y - 3z &= 0 \\ 3x + 2y + 4z &= 0 \end{aligned}$

**9.** For what value of $k$ will the following set of planes intersect in a line?

$$\begin{aligned} x - 2y - z &= 0 \\ x + 9y - 5z &= 0 \\ kx - y + z &= 0 \end{aligned}$$

## Answers

**7.** $\left(\frac{27}{4}, \frac{-15}{4}, \frac{-25}{4}\right)$ **8. a.** unique solution point $(x, y, z) = (2, 3, 4)$ **b.** no solution, 3 distinct parallel planes **c.** infinite number of solutions, the planes intersect in the line with equation $(x, y, z) = (7 - t, t, 2)$ **d.** Infinite number of solutions, the 3 planes are coincident, $x - 2y - 3z = 1$ **e.** no solution, 2 of the planes are parallel and distinct **f.** no solution, 2 planes are coincident, and the third is parallel and distinct **g.** infinite number of solutions intersecting in the line $(x, y, z) = (6 + t, -1 - t, 2t)$ **h.** no solution, planes form a triangular prism **i.** unique solution, the origin $(0, 0, 0)$ **9.** $\frac{-7}{19}$